

AI in big pharma

Analyzing the partnership between Novo Nordisk and OpenAI

Exam paper

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1. Introduction

This paper will set the scene for a discussion around how the full fledged introduction of AI in the pharmaceutical giant Novo Nordisk, can impact the company strategically. The synopsis will seek to explore broadly, through course materials, how Novo Nordisk should seek to utilize AI and how this might impact the organization. Firstly the report will seek to clarify the case and the intentions behind the partnership. Then the report will introduce a range of theoretical perspectives relevant to analyzing the case.

1.1 The case and its relevance

Ever since OpenAI on November 30th of 2022 released the first version of ChatGPT to the world, quickly breaking the record for fastest growing consumer app in history, companies have been eager to utilize the technology in order to ride the wave of technological disruption that would seemingly inevitably sweep the world and forever change the way we use technology, leading to great fortune for those able to ride the wave and great misfortune for those left at the station.(Editors, 2025)

Novo Nordisk, one of the largest pharmaceutical companies in the world has seen their stock plummet since they failed to effectively grasp the potential of their ground breaking weight loss drug Wegovy, which became a huge success, making the company the most valuable in Europe. But the potential quickly diminished as the company was unable to meet the demand, and copy cat products quickly jumped in to fill the gap in supply. In the fall of 2025 this lead to the company announcing layoffs of about 9000 employees worldwide.(*News Details*, n.d.-a) This has put the company in a strange place, they have a popular product with huge demands that they are unable to meet meanwhile they have downsized the staff significantly. In order to simultaneously increase capacity while working with a reduced staff, they are going to have to invest in becoming more efficient.

On the 14th of April 2026 Novo Nordisk releases a press release titled “Novo Nordisk and OpenAI partner to transform how medicines are discovered and delivered”.(*News Details*, n.d.-b) In the press release Novo Nordisk describes how this partnership will seek to apply advanced AI capabilities to analyse complex datasets, identify promising drug candidates, and reduce the time required to move from research to patient. OpenAI will assist Novo Nordisk in upskilling the

company's workforce and enhance AI literacy. They will seek to apply OpenAI's capabilities to improve efficiency in manufacturing, supply chain and distribution and corporate operations.

2. How AI Can Transform Novo Nordisk

2.1 The Changing Role of IT in R&D

Traditionally, R&D and IT operated as distinct silos within pharmaceutical companies. R&D created products, while IT maintained infrastructure and supported tools like ERP and CRM systems. The Novo Nordisk and OpenAI partnership fundamentally disrupts this division. As AI capabilities are embedded into drug discovery pipelines, Novo Nordisk's IT function must assume a far more central role in what the company creates.

This shift mirrors what Heppelmann describes in the context of smart, connected products, when technology becomes integral to the product itself, IT can no longer operate in the background. (Heppelmann, n.d., p. 109) For Novo Nordisk, applying OpenAI's capabilities to analyze complex datasets and identify drug candidates means that software and algorithms enter the critical path of value delivery. The organizational implication is significant, R&D and IT must now co-design processes and capabilities, dissolving the historical boundary between them.

2.2 Building an AI Factory

Iansiti and Lakhani argue that the core of a modern digital firm is a scalable decision engine, what they term an AI factory. (Iansiti & Lakhani, 2021, Chapter 3) For Novo Nordisk, the partnership with OpenAI represents the beginning of this transition. Rather than relying on individual researchers to interpret experimental results, Novo Nordisk can industrialize this process. Systematically feeding clinical trial data, patient outcomes, genomic datasets, and manufacturing metrics into a centralized AI pipeline that continuously improves its predictions.

The AI factory concept breaks down into four core components: a data pipeline, algorithm development, an experimentation platform, and software infrastructure. For Novo Nordisk, the data pipeline would aggregate information from R&D experiments, supply chain operations, and patient interactions. Algorithms would be developed and refined to predict promising drug candidates and optimize manufacturing processes. The experimentation platform would enable the company to test hypotheses at scale, something previously impossible given the time and cost of physical trials.

Finally, the software infrastructure ties these components into a coherent, interoperable system. The OpenAI partnership provides access to frontier model capabilities that can accelerate this build-out, particularly in the early stages before Novo Nordisk has developed proprietary models.

2.3 Becoming an AI Company: Five Principles

Iansiti and Lakhani outline five principles for transforming a traditional organisation into an AI company.(Iansiti & Lakhani, 2021, Chapter 5) Applied to Novo Nordisk, these principles offer a practical roadmap for the partnership with OpenAI.

One strategy demands that transformation be pursued as a unified, company-wide commitment rather than a departmental initiative. Novo Nordisk's announcement frames AI as central to their recovery and growth strategy, not a pilot project. Given the simultaneous pressures of workforce reduction and supply constraints, strategic clarity will be critical to prevent fragmented implementation across divisions.

Architectural clarity requires Novo Nordisk to consolidate its data assets across research, manufacturing, supply chain, and distribution into an integrated platform. Historically siloed datasets, from laboratory systems to ERP platforms, must be normalized and made accessible to AI models.

An agile, product-focused organization implies that Novo Nordisk must restructure how teams develop and deploy AI applications. Small, cross-functional teams, combining domain expertise in pharmacology with data science and software engineering, should be empowered to iterate rapidly on AI-enabled workflows, from drug candidate ranking to logistics optimization.

Capability foundations are addressed directly in the partnership, with OpenAI committed to upskilling Novo Nordisk's workforce and enhancing AI literacy across the organization. This is particularly relevant given the recent wave of layoffs, the remaining workforce must be equipped to work alongside AI systems rather than be displaced by them.

Clear, multidisciplinary governance will be essential given the sensitivity of pharmaceutical data and the high stakes of drug development decisions. Governance structures must address data privacy, regulatory compliance, and algorithmic accountability, especially as AI models begin to influence which compounds move into clinical trials.

3. Impact on Strategy

3.1 Evolutionary or Positioning Strategy?

When analyzing the partnership through Gavetti and Rivkins (Gavetti & Rivkin, 2007) two lenses of search for strategy, partnership appears, at first glance, primarily evolutionary. The stated focus is on enhancing employees' daily use of AI, improving efficiency in existing processes such as manufacturing, supply chain, and corporate operations. The AI capabilities are intended to make current activities faster, cheaper, and more scalable.

However, there is a latent positioning dimension that should not be overlooked. If Novo Nordisk successfully builds an AI factory capable of accelerating drug discovery, reducing the time from research to patient, this could represent a genuinely different competitive activity from rivals still relying on traditional research timelines. The company would not merely be doing the same things better, it would be doing things that slower competitors simply cannot do. The strategic ambition of the partnership, then, points toward positioning.

3.2 AI as an Organising Capability

According to Stelmaszak et al. there are three properties that characterize AI as an organizing capability: connectivity, codependence, and emergence.(Stelmaszak et al., 2026) For Novo Nordisk, connectivity could mean bridging research labs, manufacturing plants, regulatory databases, and patient outcomes into a coherent information system, something that neither human analysts nor isolated software tools could achieve at the required scale.

Codependence highlights that AI systems and human expertise are mutually reliant. Novo Nordisk's scientists provide the domain knowledge needed to train and validate models. The models, in turn, extend the scientists' analytical reach. The OpenAI upskilling initiative reflects an acknowledgement of this interdependency, the technology's value is contingent on the workforce's ability to engage with it meaningfully.

For Novo Nordisk, emergence might manifest as unexpected patterns in clinical data that lead to new therapeutic indications, or supply chain optimizations that reveal previously invisible bottlenecks. Treating AI as an organizing capability means remaining open to these emergent outcomes rather than constraining AI's role to predefined tasks.

4. Economic Implications and Ethical Considerations

4.1 Transaction Costs and Digitisation

The Novo Nordisk-OpenAI partnership aligns with Nagle, Seamans, and Tadelis' (Nagle et al., 2025) insights on digital transaction costs. Internally, AI automates complex data analysis reducing coordination costs and enabling faster decision-making. For example, AI-driven demand forecasting could help Novo Nordisk avoid the supply shortages that plagued Wegovy, lowering the transaction costs of scaling production.

Externally, digital trace data such as real-time manufacturing metrics shared with suppliers reduces asymmetric information, fostering trust and lowering contracting costs. However, the asset specificity of AI models trained on Novo Nordisk's proprietary datasets creates high switching costs, risking dependency on OpenAI.

4.2 Ethical Challenges

Algorithmic bias in drug discovery models poses a significant risk. If training datasets underrepresent certain patient populations, by age, ethnicity, or comorbidity profile, AI-generated insights may systematically overlook therapies or dosing regimens relevant to those groups.

Data security is equally critical. Clinical trial data, patient records, and proprietary compound libraries represent extraordinarily sensitive assets. Sharing data with a third-party AI provider, even one of OpenAI's caliber, introduces questions of data sovereignty and breach risk. Robust governance frameworks, clear contractual boundaries, and rigorous access controls are prerequisites for the partnership to function safely.

Finally, as AI increasingly shapes which drug candidates advance to clinical trials, questions of accountability arise. When a model recommends deprioritizing a compound that would have succeeded, or elevating one that fails, the organization must be able to explain and defend those decisions to regulators, patients, and shareholders.

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